POST LAB TASKS

COURSE CODE: C1002

SUBJECT: PROGRAMMING FUNDAMENTALS

INSTRUCTOR: MR. SHAYAN ALI

NAME: AMNA KHAN

ROLL NO/ID: 26K-3091

CLASS/SECTION: C

DATE OF SUBMISSION: 18-OCT'2026

Question 1: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

1

1 1

1 1 1

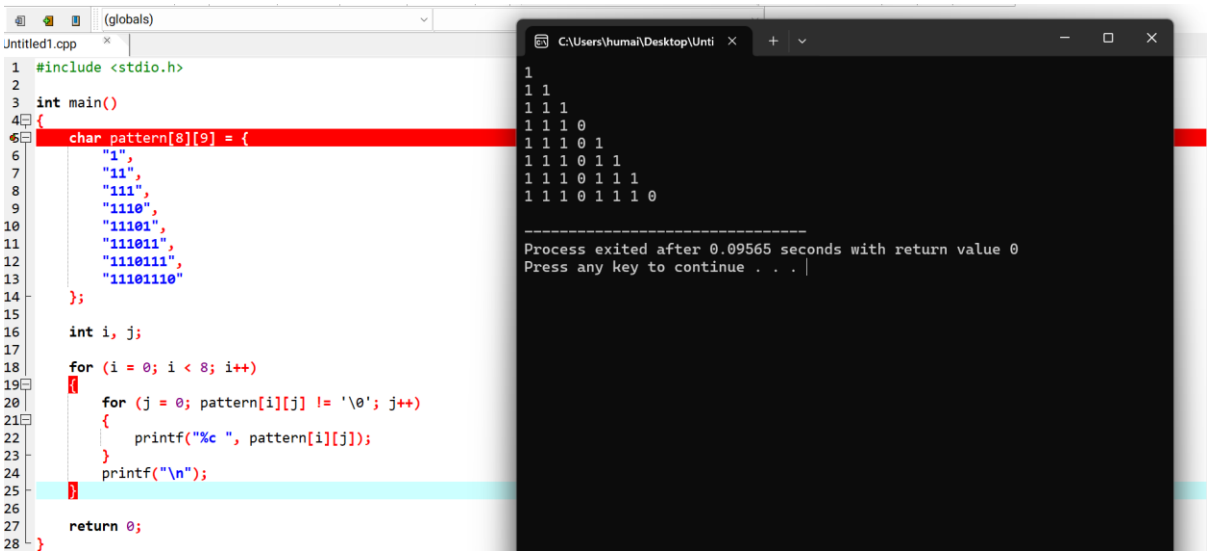
1 1 1 0

1 1 1 0 1

1 1 1 0 1 1

1 1 1 0 1 1 1

1 1 1 0 1 1 1 0



Question 2: Print the Following Pattern

Write a C program using nested loops to print the following pattern:

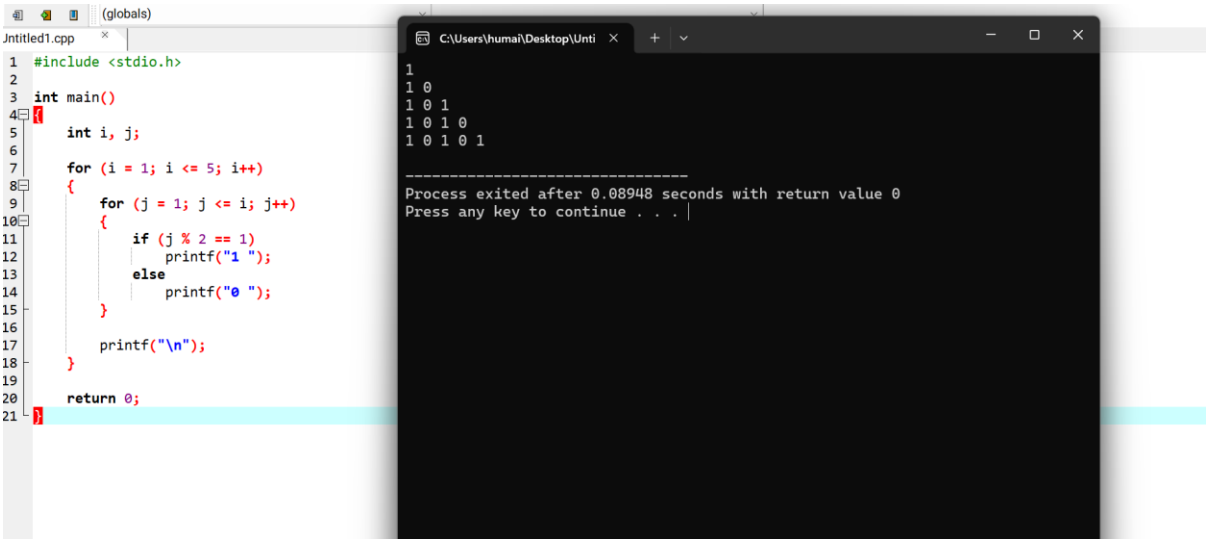
1

1 0

1 0 1

1 0 1 0

1 0 1 0 1



Question 3: Matrix Addition (2D Arrays)

Write a C program to input two 3×3 matrices using separate 2D arrays, add their corresponding elements using nested loops, store the results in a third matrix, and display both input matrices and the resulting sum matrix

(globals)

[*] Untitled1.cpp

```c
#include <stdio.h>

int main()
{
    int a[3][3], b[3][3], c[3][3];
    int i, j;

    printf("Enter elements of first matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter elements of second matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
```

[*] Untitled1.cpp

```c
    }

    printf("Enter elements of second matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            c[i][j] = a[i][j] + b[i][j];
        }
    }

    printf("\nFirst Matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            printf("%d\t", a[i][j]);
        }
        printf("\n");
    }

    printf("\nSecond Matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
```

[*] Untitled1.cpp

```c
    }

    printf("\nFirst Matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            printf("%d\t", a[i][j]);
        }
        printf("\n");
    }

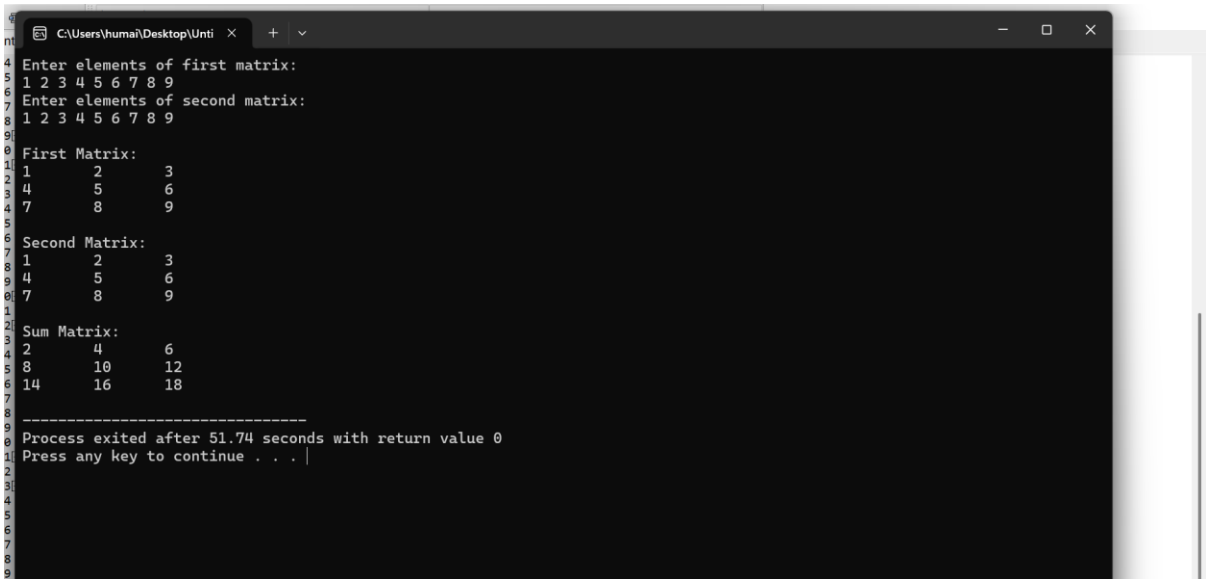
    printf("\nSecond Matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            printf("%d\t", b[i][j]);
        }
        printf("\n");
    }

    printf("\nSum Matrix:\n");

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            printf("%d\t", c[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```



Question 4: Student Marks Analysis (2D Array)

Write a C program using a 2D array to store the Mid and Final exam marks of 4 students (each row is a student, with one column for Mid and one for Final). Input the marks using nested loops, then calculate and display:

1. The total marks of each student (Mid + Final)

2. The highest marks in Mid, the highest marks in Final, and the highest marks overall across both

3. The lowest marks in Mid, the lowest marks in Final, and the lowest marks overall across both

4. The average of Mid marks and the average of Final marks, calculated separately

(globals)

[*] Untitled1.cpp

```c
#include <stdio.h>

int main()
{
    int marks[4][2];
    int i, j, total;
    int highMid, lowMid, highFinal, lowFinal;
    int highOverall, lowOverall;
    int sumMid = 0, sumFinal = 0;

    printf("Enter Mid and Final marks of 4 students:\n");

    for (i = 0; i < 4; i++)
    {
        printf("Student %d: ", i + 1);

        for (j = 0; j < 2; j++)
        {
            scanf("%d", &marks[i][j]);
        }
    }

    highMid = lowMid = marks[0][0];
    highFinal = lowFinal = marks[0][1];

    highOverall = lowOverall = marks[0][0];

    for (i = 0; i < 4; i++)
    {
        total = marks[i][0] + marks[i][1];
```

(globals)

[*] Untitled1.cpp

```c
    for (i = 0; i < 4; i++)
    {
        total = marks[i][0] + marks[i][1];

        printf("Student %d total = %d\n", i + 1, total);

        sumMid += marks[i][0];
        sumFinal += marks[i][1];

        if (marks[i][0] > highMid)
            highMid = marks[i][0];

        if (marks[i][0] < lowMid)
            lowMid = marks[i][0];

        if (marks[i][1] > highFinal)
            highFinal = marks[i][1];

        if (marks[i][1] < lowFinal)
            lowFinal = marks[i][1];

        for (j = 0; j < 2; j++)
        {
            if (marks[i][j] > highOverall)
                highOverall = marks[i][j];

            if (marks[i][j] < lowOverall)
                lowOverall = marks[i][j];
        }
    }
```

[*] Untitled1.cpp

```c
        if (marks[i][1] > highFinal)
            highFinal = marks[i][1];

        if (marks[i][1] < lowFinal)
            lowFinal = marks[i][1];

        for (j = 0; j < 2; j++)
        {
            if (marks[i][j] > highOverall)
                highOverall = marks[i][j];

            if (marks[i][j] < lowOverall)
                lowOverall = marks[i][j];
        }
    }

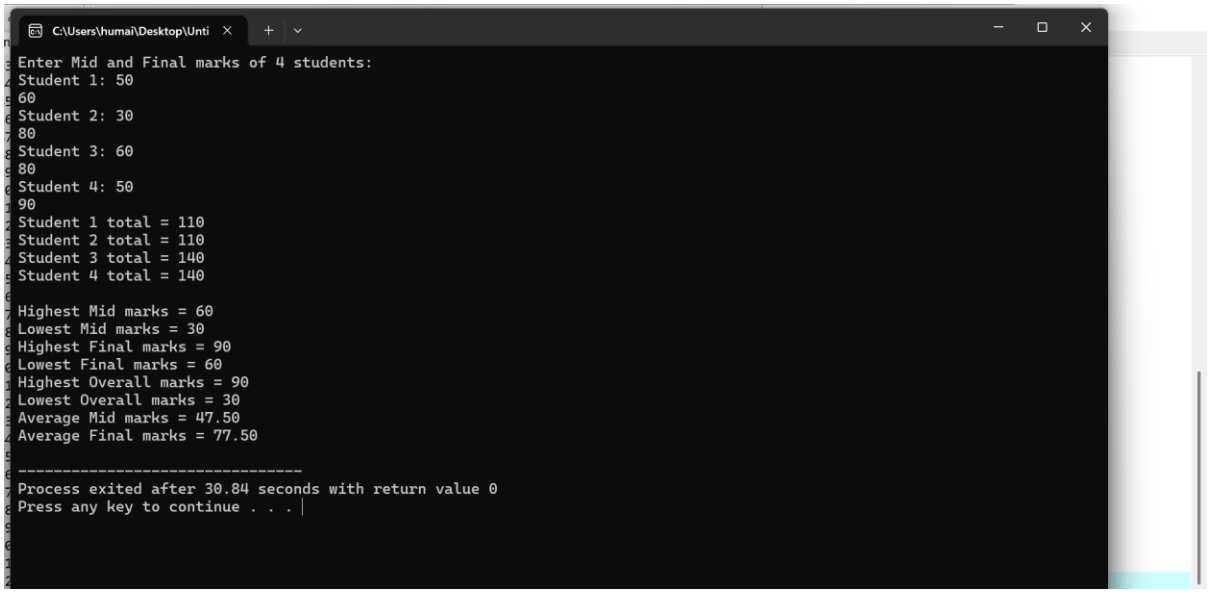
    printf("\nHighest Mid marks = %d\n", highMid);
    printf("Lowest Mid marks = %d\n", lowMid);

    printf("Highest Final marks = %d\n", highFinal);
    printf("Lowest Final marks = %d\n", lowFinal);

    printf("Highest Overall marks = %d\n", highOverall);
    printf("Lowest Overall marks = %d\n", lowOverall);

    printf("Average Mid marks = %.2f\n", sumMid / 4.0);
    printf("Average Final marks = %.2f\n", sumFinal / 4.0);

    return 0;
}
```



Question 5: 3D Array Scenario (Paper Work)

A university stores the marks of students in a 3D array int marks[2][3][3], where the first index is the semester (2 semesters), the second index is the student (3 students) and the third index is the subject (3 subjects). On paper, draw the array as 2 layers (one per semester) and label every cell with its index marks[semester][student][subject]. Decide what to insert: fill every cell with a realistic mark of your choice (0-100), writing the name of the semester, each student and each subject on the margins of your diagram. Then, from your diagram, write down: (a) the index and value of the highest mark, (b) the marks of student 1 in semester 2 across all subjects, and (c) the total marks of each student in semester 1.

Date:

int marks [2][3][3];

- 2 semesters
- 3 students
- 3 subjects

Semester 1 - marks [0]

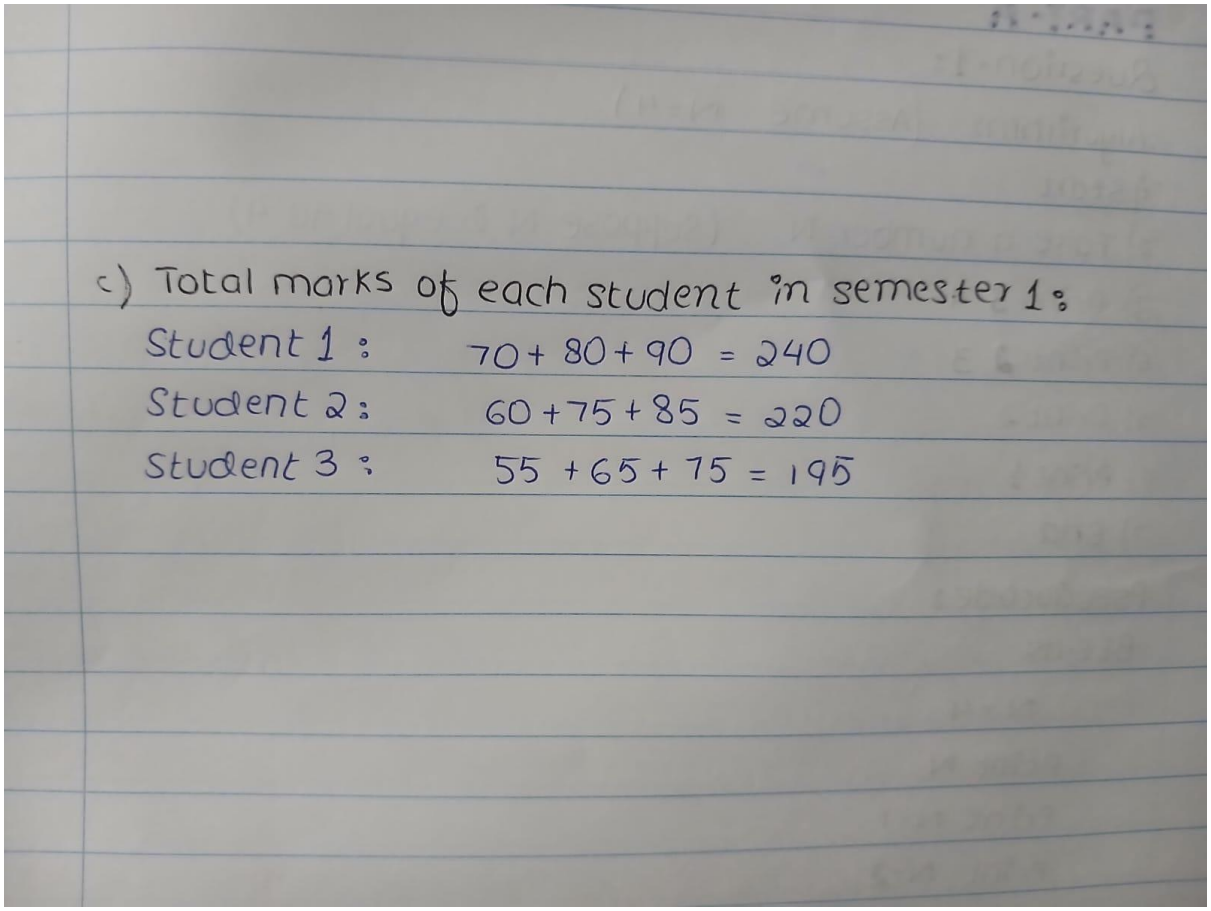
Student	Subject 1	Subject 2	Subject 3
Student 1	70	80	90
Student 2	60	75	85
Student 3	55	65	75

Semester 2 - marks[1]

Student	subject 1	subject 2	Subject 3
Student 1	88	92	78
Student 2	72	81	90
Student 3	68	74	82

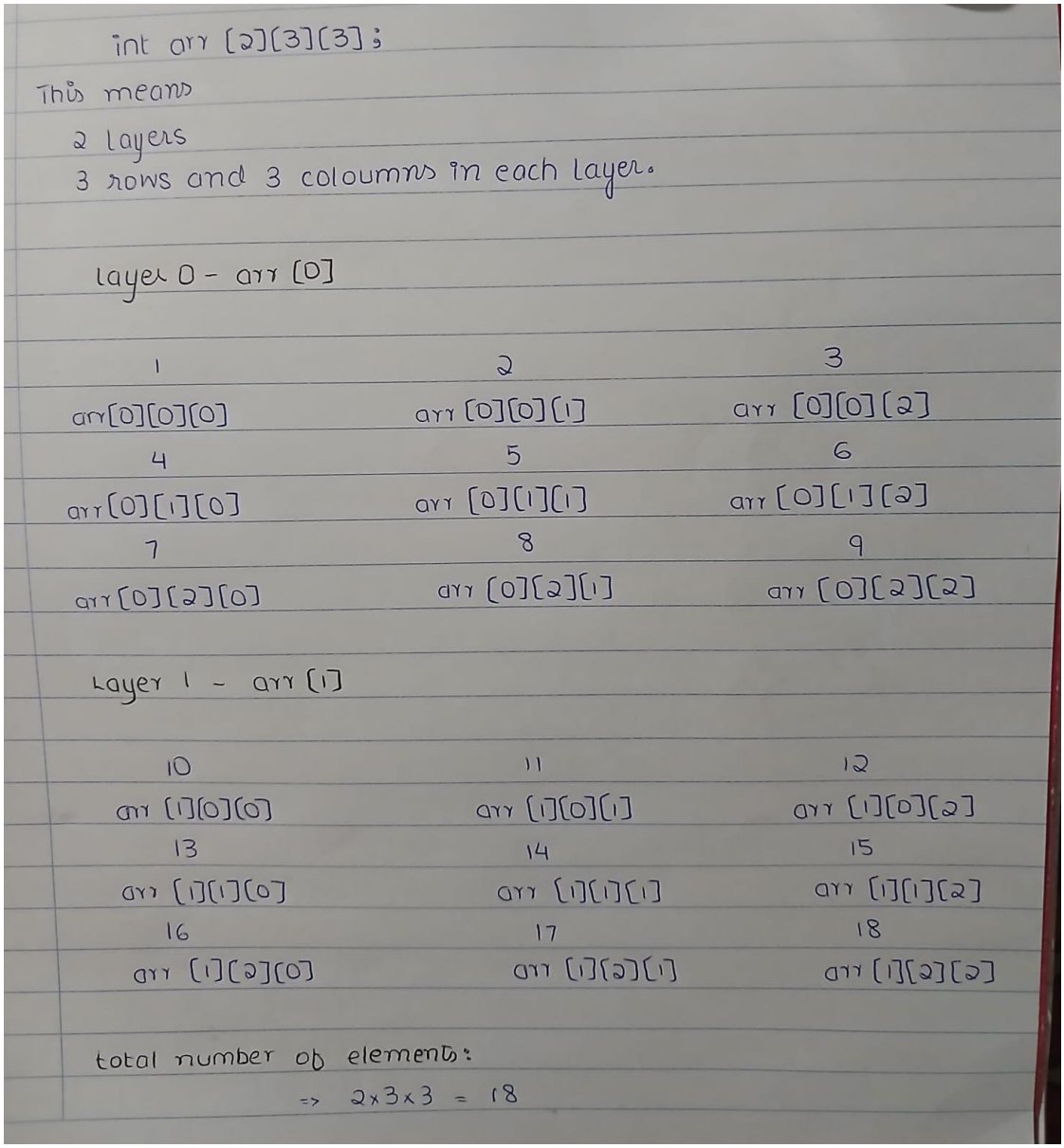
a) Highest marks and index:
The highest mark is 92 and its array index is
marks [1][0][1] = 92;

b) marks of student 1 in semester 2:
marks [1][0][0] = 88;
marks [1][0][1] = 92;
marks [1][0][2] = 78;



Question 6: 3D Array on Paper (Paper Work)

On paper, draw the 3D array int arr[2][3][3] as 2 layers (pages), each layer containing a 3×3 grid. Label every cell with its index in the form arr[i][j][k] and label the layers, rows and columns. Fill the cells with any values of your choice.



Question 7: 3D Array on Paper (Paper Work)

On paper, draw the 3D array int arr[3][2][4] as 3 layers (pages), each layer containing 2 rows and 4 columns. Show the index of every cell, fill the array with values, and then write the index of the first and last element of each layer and the total number of elements in the array.

int arr [3][2][4];

Layer 0 - arr [0]

1	2	3	4
[0][0][0]	[0][0][1]	[0][0][2]	[0][0][3]
5	6	7	8
[0][1][0]	[0][1][1]	[0][1][2]	[0][1][3]

first element: arr [0][0][0] = 1
last element: arr [0][1][3] = 8

Layer 1 - arr [1]

9	10	11	12
[1][0][0]	[1][0][1]	[1][0][2]	[1][0][3]
13	14	15	16
[1][1][0]	[1][1][1]	[1][1][2]	[1][1][3]

first element: arr [1][0][0] = 9
Last element: arr [1][1][3] = 16

Layer 2 - arr[2]

17	18	19	20
[2][0][0]	[2][0][1]	[2][0][2]	[2][0][3]
21	22	23	24
[2][1][0]	[2][1][1]	[2][1][2]	[2][1][3]

first element: arr [2][0][0] = 17

last element: arr [2][1][3] = 24

Total number of elements:

$3 \times 2 \times 4 = 24$